«Approved»
By Director, School of Computer Engineering
Prof. Praveen Kumar
_____
«___» of _____ 2025

Syllabus
Academic Year 2025-2026

1. General information	
Course Code	
Course Title	Advanced Programming (Python)
Degree Cycle (Level)/ Major / Relation to curriculum	Bachelor's Degree 6B06301 - Cybersecurity
Year, trimester	2rd year, 5th term.
Language of Instruction:	English
Lecturer(s)/ Instructors/ Instructor Contact Information	1. Islam Omirzak, M.Sc. Ed, PhD Candidate in Computer Science, SSE, Senior Lecturer, i.omirzak@astanait.edu.kz, Astana IT University, Expo, C3 block, 3rd floor, C1.3.354; Office hours: Wednesday, 6 p.m., C1.3.354 2. Turysbek Nazym, MSc in Advanced Computing , SSE, Lecturer., Astana IT University, Expo, C3 block, 3rd floor, Office hours: Friday, 6 p.m., C1.3.354 3. Kenzhegali Nurgaliyev, MSc & PhD Candidate in Information Systems, Teacher, kenzhegali.nurgaliyev@astanait.edu.kz, Astana IT University, Expo, C3 block, 3rd floor. Office hours: Wednesday, 3 p.m., C1.3.354
Number of Credits	5
Workload of course components and credits per trimester	

ECTS credits	Contact hours			TSIS	SIS	Total hours
	Lectures	Practice sessions	Laboratory work			
5	20	20	10	10	90	150

Prerequisites	Introduction to Programming, Algorithms & Data structures, OOP
Post requisites	Advanced Programming II, Machine Learning, Data Analysis
2. Goals, objectives and learning outcomes of the course	
1. Course Overview/Description	This course is aimed to introduce students to the Python programming language and its libraries. A structure of the course is oriented on procedural programming, algorithm designing, working application forms (libraries), object-oriented programming, asynchronous programming, creating database applications. During the process students will also gain knowledge of creation of web apps and RESTful APIs and use of ORM library (SQLAlchemy or similar). Additionally, this course

	provides students understanding of using non-strict types of variables. Upon completion, students should be able to code, test, create and debug fully functional and complicated applications on Python programming language.
2. Course Learning Goals	Course goal is to provide a solid background in developing RESTful API with Python for students of Software Engineering, in preparation either for a job in industry or for more advanced courses at the graduate level.
3. Course Learning Outcomes	By the end of this course the students will be able to: - Display understanding of programming paradigms in Python. - Demonstrate knowledge of how Python code is built and executed. - Apply obtained skills to write truly well-designed and stable software in Python. - Design a flexible architecture that serves as a program module or package
4. Methods/forms of teaching	- Quiz. - Practical assessments.
5. Coursework and Grading Scheme	Midterm, Endterm, Labs, Project
6. Academic Integrity	Students are expected to uphold the highest standards of academic honesty. All submitted work must be the result of the student's own effort unless group work is explicitly allowed. Plagiarism, cheating on exams, unauthorized collaboration, or copying code from online sources without proper citation are considered violations of academic integrity. Any breach of academic integrity will result in penalties, which may include: • Reduction of grade or a zero for the assignment/exam. • Reporting to the university's disciplinary committee. • Possible failure of the course. Maintaining academic integrity is essential for developing genuine knowledge and professional skills
7. Learning resources:	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). **Basic literature:** • www.python.org – Python official web site with fully provided documentation. [Online] • https://docs.sqlalchemy.org/en/14/ – SQLAlchemy ORM official web site with fully provided documentation. [Online] • www.fastapi.tiangolo.com – FastAPI. official web site with fully provided documentation. [Online] • https://pandas.pydata.org/docs/ – Pandas official web site with fully provided documentation. [Online] • https://flask.palletsprojects.com/en/2.2.x/ – Flask web framework official web site with fully provided documentation. [Online] • Mueller, J.P. (2018). *Beginning Programming with Python for dummies.* 2nd editions. https://www.pdfdrive.com/beginning-programming-with-python-for-dummies-e176211134.html **Supplementary literature:** Cisco Corp., *Programming Essentials in Python, 2018*, www.netacad.com Edube[Online] Matthes E. Python Crash Course: A Hands-on, Project-based Introduction to Programming 3rd edition, No Starch Press, 2023

8. Technology employed:	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
9. Course Policies	Course Policies for In-Person Instruction: **Course Policies for In-Person Instruction:** **Presence and Participation:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. **Deadlines and Late Submissions:** Late submissions and/or defenses will not be accepted. Any assignment submitted after the specified deadline will receive zero points. Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **The Honor Code:** Class meetings and sections in this course will be highly interactive, you will hear plenty of great ideas from your peers, and if things go as planned, you might even change your mind based on what others say! However, you will be evaluated in this course as an individual and are expected to cite sources and individuals from whom you have learned and borrowed as a display of academic, intellectual, and creative integrity. In addition, while we encourage collaboration during class activities and assignments as an aid to your learning (with collaborators and their contributions cited in submissions), we expect you to complete the three exams in the course unaided, to demonstrate your proficiency with course material. This means conversations about exam questions, and consultation of course or other resources while working on exams questions, are not permitted. Failure to abide by these expectations is a violation of Stanford's Honor Code and is a serious offense, even when the violation is unintentional. Conduct prohibited by the Honor Code includes all forms of academic dishonesty, among them unpermitted collaboration and representing others' work as one's own. Please review Stanford's Honor Code, and documentation and citation resources from the Hume Center for Writing and Speaking. When in doubt, contact your section leader. **Course Policies for Use of Generative AI:** Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams;

2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

These expectations apply to **AI-generated content** as well. When using generative AI tools:

1. Validate the accuracy of any AI-generated content with at least one reliable reference, because generative AI can produce questionable and biased results, including misinformation and "hallucinations".
2. Properly attribute your use of AI tools in accordance with the class citation requirements.
    a. Do not share any personal identifiable information (PII) or other sensitive information in your prompts to an AI platform, because generative AI tools may store your prompts (review the AI tool's "terms of use" regarding data privacy).
3. Revise any AI-generated output to create authentic content that reflects your unique experience with the course material.

Extended Absence

In cases of documented exceptional circumstances, such as medical emergencies or other valid reasons, students may be granted extensions or alternate arrangements for assignment defenses. However, these circumstances must be communicated to the instructor well in advance whenever possible.

Late Homework Policy: The assignments will be evaluated only if the student has uploaded his/her work to the moodle and defended it on the due date. Without defense, a student cannot get a grade for assignments. . You are permitted 3 "late days" for your assignments.

- 1 day late: 25% of the maximum allowable marks
- 2 days late: 50% of the maximum allowable marks
- 3 days late: 75% of the maximum allowable marks

Course Privacy Statement

As noted in the university's recording and broadcasting courses policy, students may not audio or video record class meetings without permission from the instructor (and guest speakers, when applicable). If the instructor grants permission or if the teaching team posts videos themselves, students may keep recordings only for personal use and may not post recordings on the Internet, or otherwise distribute them. These policies protect the privacy rights of instructors and students, and the intellectual property and other rights of the university. Students who need lectures recorded for the purposes of an academic accommodation should contact the Office of Accessible Education.

Exam Policy:

Assignments are designed to reinforce the understanding of course materials and prepare students for the midterm and endterm exams. If a student fails to defend any/all assignments, he may not be permitted to sit for the midterm or endterm exams. This is because the content and concepts covered in the exams are directly

	related to the assignments and their defenses. Attestation I and II**:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Final exam** for the course final project defense and final project documentation submission.
10. Course Schedule	The official course schedule, including lecture and practical session times, is published on the university's system at **https://du.astanait.edu.kz**. Students are required to regularly check the system for updates. Any changes to the schedule (such as room changes, rescheduled classes, or instructor substitutions) will be reflected there. The course instructor is not responsible for individually notifying students of schedule updates. Attendance requirements apply according to the updated schedule as published in the system.
11. Syllabus Inclusion Statements	Our classroom is a respectful and inclusive environment. Diversity of backgrounds, experiences, and perspectives is valued and encouraged. If you have any accessibility needs, please contact me or the university's disability support services so that appropriate accommodations can be arranged.

3.1 Abbreviations

#	Abbreviation	Meaning
1	IASS	Instructor assisted student study - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
2	SIS	Student independent study
3	IW	Individual work
4	GW	Group work
5	PA	Practical assignment
6	LW	Laboratory work
7	MCQ	Multiple choice quiz
8	PBT	Paper-based test

NB: You may introduce abbreviations that are used in the syllabus

3.2 Course Plan

Week No	Course Topic	Lectures (h/w)	Practice sessions (h/w)	Lab. Sessions (h/w)	TSIS (h/w)	SIS (h/w)

1	Introduction to Python Programming: ● Review, history, goals and versions ● Python3 installation ● Programming Environments ● Built-in instruments and pip ● Useful links and official documentation ● First program creation and running on different OSs ● Python print() function ● Short description of python programming components	2	3	0	1	9
2	Data types and operations over them: ● Identifiers and keywords ● Numbers ● Strings o Indexing o Slicing o Comparison o Formatting o Methods o Encoding Working with collections: ● Sequences o Tuples o Named tuples o Lists ● Sets o Set o Frozenset ● Data Mapping o Dictionary o Dictionary with default values ● Loop avoidance and collections copying o Iterators, functions and operators to work with iterating objects o Collection copying	2	3	0	1	9

3	Controlling structures and functions: ● Conditionals and Loops ● Error (exception) handling & own exceptions ● Functions o Function names and documenting o Document and parameters extraction o Recursion o Nested functions o Global variable access o Lambda functions o Asserts ● Asynchronous Programming	2	3	0	1	9
4	Working with modules and files: ● Reading and writing of binary files ● Writing and analysis of txt files ● Context manager “with” ● Working with JSON files Object-oriented programming in Python: ● OOP principles ● Own classes ● Dynamic code execution and dynamic import ● Decorators and annotations of functions and methods ● Functors ● Descriptors ● Class decorators	2	3	0	1	9
5	Working with databases and network interaction: ● Databases in python SQLite3	2	3	0	1	9

6	Django Web Application Development ● Environment setup & installation ● REST API ● Postman for testing ● Endpoint implementation	2	3	0	1	9
7	Introduction to Data Science/Jupyter ● Introduction to Jupyter notebook ● Variables in Jupyter ● Jupyter Lab	2	3	0	1	9
8	Intro to numpy • NumPy arrays, operations, and statistics	2	3	0	1	9
9	Intro to Pandas ● DataFrames, Series	2	3	0	1	9
10	• Into to Matplotlib	2	3	0	1	9
	Total hours: 150	**30**	**20**	**0**	**10**	**90**

3.3 Detailed course plan

Week No	Lecture Topics with Learning Resources	Tasks and Activities for Practice Sessions with Recommended Reading Resources
1	Introduction & Basics	Write a "Hello, Python!" program that takes user input and prints a custom message. • Solve simple problems: converting temperature, basic calculator, even/odd check.
2	Data Types & Collections	• String manipulations (reverse, count letters, slicing). • Create a contact list using lists and dictionaries. • Practice with tuple unpacking and set operations.
3	Control Structures & Functions	• Implement factorial, Fibonacci, and prime number check using loops and recursion. • Write a program that validates passwords using functions. • Exception handling tasks.
4	Modules, Files, & OOP	• File operations: read/write CSV, JSON, and text. • Build your own module (e.g., math_utils.py). • Create a BankAccount class with deposit, withdraw, and balance.
5	Databases (SQLite3)	Create a SQLite3 database for storing user info (CRUD operations).

6	Django Web Application Development	Environment setup, Django models/serializers, implementing **REST API** endpoints (GET, POST, PUT, DELETE), using **Postman for testing**.
7	Intro to Data Science/Jupyter	• Set up Jupyter Notebook and document past Python scripts. • Simple data visualization in notebook (text or print output).
8	Intro to numpy	• Solve numeric problems using NumPy arrays. • Matrix operations (addition, multiplication).
9	Intro to Pandas	Load CSV data into a DataFrame, clean missing data, compute summary stats.
10	Into to Matplotlib	Plot various charts (bar, line, scatter, histogram).

3.4 List of assignments for Student Independent Study

№	Assignments (topics) for Independent Study	Hours	Recommended literature and other sources (links)	Form of submission and deadlines
1	2	3	4	5
1	Lab 1 – Mini tasks: calculator, area of a circle, temperature converter	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Lab 2 – Contact Book Manager	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Lab 3 – Advanced Calculator with Error Handling	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	Lab 4 – JSON-Based Inventory System	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
5	Lab 5 – Build a “Student Database” app using SQLite3 with basic CLI menu	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	Lab 6 – Build a Basic CRUD REST API (e.g., Simple To-Do List)	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
7	Lab 7 - Working with Jupyter Notebook	3	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
8	Lab 8 - Lab: create a mini simulation (e.g., dice rolls or	3	Reading list, books, internet resources	Reading list, books, internet resources Electronic version

	weather data generator).		Electronic version	

3.5 For instructor assisted student study (TSIS) schedule and policy follow your course LMS Moodle.

4. Student performance evaluation system for the course.

Period	Assignments*	Number of points	Form of assessment	Total
1st attestation	Assignment 1. Setting up Programming environment	**15**	Practical task, oral defense, in practice session time, quizzes, submission in https://lms.astanait.edu.kz	100
	Assignment 2. Data types, Lists	**15**		
	Assignment 3. Decision statements & Classes in Python	**15**		
	Assignment 4. Testing your code	**15**		
	Midterm	**40**		
2nd attestation	Assignment 5. Web Application Development with Flask/Django	15	Practical task, oral defense, in practice session time, quizzes, submission in https://lms.astanait.edu.kz	100
	Assignment 6. Working with APIs	15		
	Assignment 7. Data Visualization and Analytics	15		
	Assignment 8. Deploying an app	15		
	End term	**40**		
Final exam	Presentation of final project	100	Oral defense with presentation, submission in https://lms.astanait.edu.kz	100
Total	**0.3*1 st Att + 0.3 *2 nd Att + 0,4 + Final**			**100**

** The complete title of the assignment must be clearly stated.*
*** The number of assignments may vary depending on the course content and is determined by the course syllabus.*
**** Description of iROS or Social GPA elements, if applicable*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	GPA	Percentage	**Description**
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	30-49	Fail
F	0	0-29	

Course Assessment Rubric for Student Performance Evaluation

Learning outcomes	Indicators	Excellent (90-100)	Good (70-89)	Satisfactory (50-69)	Fail 0-49
1.Student demonstrates comprehensive understanding of course content	1.1.Accurately applies theoretical and practical concepts. 1.2 Shows consistency across all assignments.	An "excellent" grade (90-100%) is given if the student has thoroughly mastered the course material and completed all assignments with strict adherence to the requirements.	A "good" rating (70-89%) is given if the student demonstrates a strong understanding of the course material and completes assignments with only minor deviations from the requirements.	A "satisfactory" grade (60-69%) is given if the student demonstrates a partial understanding of the course material and completes assignments that meet only some of the requirements.	A grade of "fail" (0–49%) is given if the student demonstrates a poor understanding of the course material and completes assignments with significant deviations from the requirements or no significant assessable material, absent or assessment missing a must pass component
2. . Student consistently follows assignment instructions and course guidelines	2.1 Meets deadlines and formatting/technical requirements. 2.2 Submissions align with task specifications.	The content of the assignments must fully align with the design concepts taught in the course.	The work closely aligns with the design concepts, showing solid proficiency in UX/UI principles.	The work produced somewhat aligns with the design concepts but may lack consistency and thoroughness. The student's design solutions are basic and may only partially address user needs.	The work produced only loosely aligns with the design concepts and often lacks coherence or relevance.
3. The	3.1 Design outputs	Additionally,	The student's	Problem-	The student's

student's work reflects key design principles taught in the course.	are aligned with taught concepts (e.g., hierarchy, balance, consistency). 3.2 Demonstrates ability to translate theory into practice.	the student must demonstrate an excellent ability to evaluate and create user-friendly, responsive, and accessible interfaces.	design solutions are functional and user-centered, effectively addressing user needs, though they may lack some innovation or polish.	solving is minimal, with simple solutions provided, often lacking creativity or depth.	design solutions are ineffective, failing to adequately address user needs.

5. Course assessment methodology

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **TSIS (Teacher Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor
- **SIS** – Student engagement in self-directed study on assigned topics and the completion of relevant assignments given in the syllabus.
- **Mid-term and End-term** are a review of the topics covered and an assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of theoretical and practical parts of the course to evaluate the students' academic performance and professional skills. At the completion of the exam, all works must be submitted in the Learning Management System (moodle.astanait.edu.kz). No late submissions are allowed in the exam.

6. Lecturer (lecturers) signature

Full name	**Job title**	**Date**	**Signature**
Islam Omirzak	Senior Lecturer		
Nazym Turysbek	Lecturer		
Kenzhegali Nurgaliyev	Teacher		